1.4 e Is Irrational, and So Is e^2

WORKSHEET

This section uses two questions. The first, from 1997, proves that e is irrational. The second, from 2024, uses a cleverer version of the same idea to prove that e^2 is irrational, which turns out to be genuinely harder.

Warm-up

STEP III 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q7

Main question

STEP III 2024, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Suppose $e = p/q$. Multiplying e by q gives an integer. What does multiplying by $q!$ do to each term of the series for e?

STAGE 1 The warm-up: e is irrational

Covers: the warm-up question, STEP III 1997

Ask yourself b_n is a geometric series. What are its first term and common ratio, and so what is its sum?

Pause and try Compare a_n and b_n term by term: each denominator in a_n is a product of factors that are all at least $n+1$. Which series is bigger?

Ask yourself Multiply the series for e by $n!$. The terms up to $1/n!$ give an integer; which series do the rest make up? If $e = p/q$, which value of n gives a contradiction?

STAGE 2 Why e^2 is harder

Covers: the step from the warm-up to the main question, STEP III 2024

Ask yourself Try the warm-up's method on $e^2 = \sum \frac{2^r}{r!}$. Why does the tail no longer get small? If $e^2 = \frac{p}{q}$, what can you say about qe and $\frac{p}{e}$?

STAGE 3 The two tails

Covers: main question, Parts (i) and (ii)

Ask yourself Part (i) is the warm-up's comparison again, with $f(n)$ in place of a_n.

Ask yourself The terms of $g(n)$ alternate in sign. Are they getting smaller?

Pause and try Group the terms in pairs, in two ways: from the first term, and from the second. Each grouping gives one of the inequalities.

STAGE 4 Integers again

Covers: main question, Part (iii)

Ask yourself Write $e = \sum_{r \geq 0} \frac{1}{r!}$ and multiply by $m!$, where $m = 2n$. Which terms become integers?

Ask yourself What do the remaining terms, those with $r > m$, add up to? Compare them with the definition of $f(m)$.

Pause and try Do the same with $\frac{1}{e} = \sum_{r \geq 0} \frac{(-1)^r}{r!}$. Why does it matter that $m = 2n$ is even when you match the tail with $g(m)$?

STAGE 5 The contradiction

Covers: main question, Parts (iv) and (v)

Ask yourself Use part (iii) to write $f(2n)$ and $g(2n)$ as integers plus or minus a multiple of e or $\frac{1}{e}$.

Pause and try Substitute these into $q\,f(2n) + p\,g(2n)$. Where does $qe = \frac{p}{e}$ come in?

Ask yourself Suppose $e^2 = \frac{p}{q}$. Then part (iv) applies for every n.

Pause and try Parts (i) and (ii) squeeze $q\,f(2n) + p\,g(2n)$ between 0 and something small. How large must n be for that integer to lie strictly between 0 and 1?

What's really going on

How irrational is e?

Joseph Liouville proved in 1840 that e^2 is irrational, and in fact that it doesn't satisfy any quadratic equation with integer coefficients. In 1873 Charles Hermite went much further: he proved that e is *transcendental*, meaning it is not a root of *any* polynomial with integer coefficients. His proof is a sophisticated version of exactly the argument above: build an integer, show it is strictly between 0 and 1. Nine years later Ferdinand von Lindemann adapted Hermite's method to show that π is transcendental, finally proving that the ancient problem of "squaring the circle" with ruler and compass is impossible.

What we still don't know

For all the power of these methods, embarrassingly simple questions remain open. Is $e+\pi$ irrational? Is $e\pi$? Nobody knows, although at least one of them must be transcendental (exercise 4). Is $\zeta(3) = 1 + \frac{1}{8} + \frac{1}{27} + \cdots$ irrational? Yes: Roger Apéry astonished the mathematical world by proving it in 1978. Is $\zeta(5)$? Nobody knows.

Approximation

Behind these proofs lies a deeper idea: a number is irrational if it can be approximated by fractions *too well*. The partial sums of the series for e are fractions that get very close to e very quickly; a fraction can't be approximated by other fractions that well. This is the starting point of *Diophantine approximation*, a branch of number theory you can explore further in Section 2.9, where a different family of integrals produces even better fractions closing in on e.

Try this next

1 **Using $1/e$ alone.** Prove that e is irrational by supposing $1/e = p/q$ and using only the alternating series for $1/e$.

2 $\sin 1$ **is irrational.** Use $\sin 1 = 1 - \dfrac{1}{3!} + \dfrac{1}{5!} - \cdots$ to prove that $\sin 1$ is irrational. (Multiply by a suitable odd factorial.)

3 **Why not π?** Explain why this method doesn't work directly for π. Then look up Ivan Niven's one-page proof (1947) that π is irrational, which uses an integral instead of a series. It is within reach of a strong A-level student.

4 **One of them is transcendental.** HARDER Show that e and π are the roots of $x^2 - (e+\pi)x + e\pi = 0$. You may assume that a root of a polynomial whose coefficients are algebraic numbers is itself algebraic, and that π is not algebraic (Lindemann). Deduce that at least one of $e+\pi$ and $e\pi$ is transcendental.

Further reading

- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 8, "Some irrational numbers". Proves that e, e^2, e^4 and π^2 are irrational, using exactly these ideas.
- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). A short classic covering everything from $\sqrt{2}$ to transcendence.
- **Julian Havil, *The Irrationals*** (Princeton University Press). A readable history of irrational numbers.

The full guided solution is in Section 1.4:
https://beyondstep.keithmcnulty.org/ch01/e-irrational.html